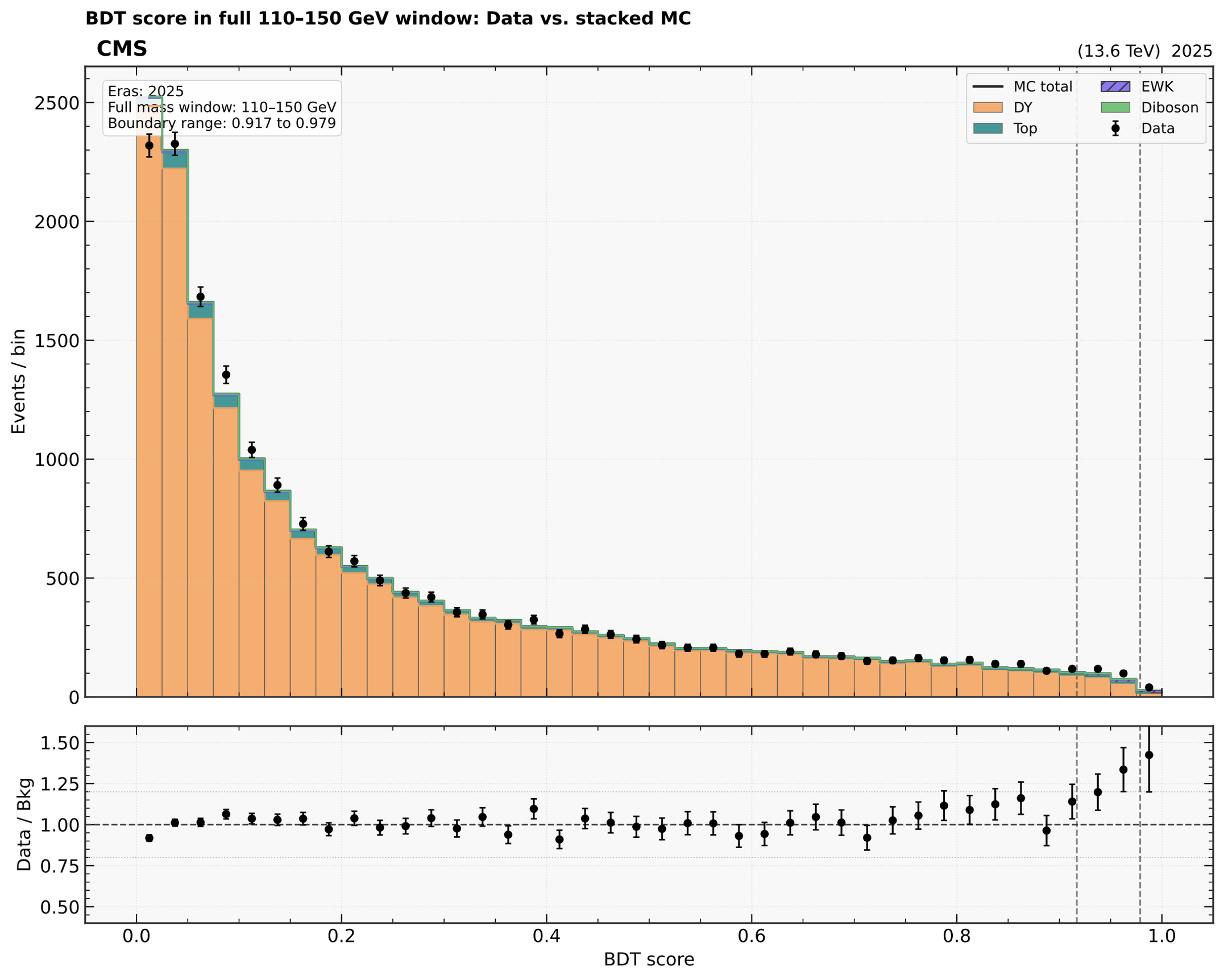
Run-3 H → μμ VBF BDT Category Optimisation

Eras : 2025
Category edges : 0.0000, 0.9170, 0.9788, 1.0000
MC scale factor : 1.0000 (Data/MC sideband normalisation)
Blinding window : 115–135 GeV
Sidebands : (110,115) ∪ (135,150) GeV
Outputs dir : /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Extra selection : –

CONTENTS:

p.2 BDT score distribution: Data vs. MC
p.3 BDT bin yields
p.4 Significance vs. number of categories
p.5 Boundary scan step-by-step (AN Fig. 32)
p.6 Atanh BDT + S/√B per category
p.7 BDT output vs. mass hypothesis (AN Fig. 26R)
p.8 Signal FWHM vs. BDT score
p.9 ROC curve with AUC
p.10 Blinded dimuon mass spectrum
p.11 Mass sculpting check (AN Fig. 25)
p.12 Per-category mass distributions and background fits
p.13 Background model comparison (χ^2/ndf)
p.14 VBF signal DCB fits per category
p.15 DCB parameter table
p.16 σ_{CB} vs. category
p.17 Signal acceptance × efficiency
p.18 Per-category yield and sensitivity summary
p.19 Signal mass shapes mH=120/125/130 (AN Fig. 40)
p.20 Category summary table
p.21 Category process composition

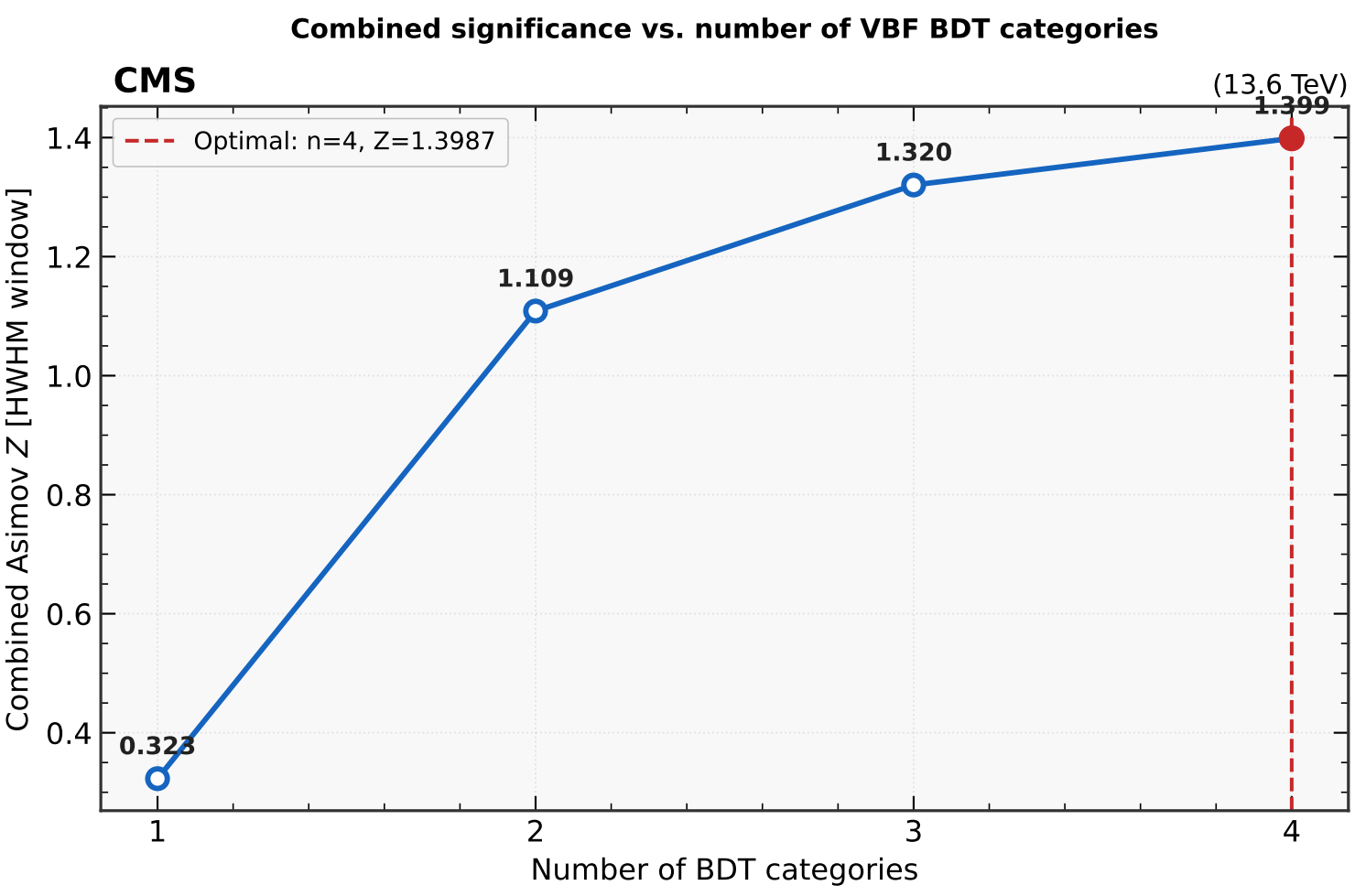

BDT score in full 110–150 GeV window: Data vs. stacked MC
CMS
(13.6 TeV) 2025
Eras: 2025
Full mass window: 110–150 GeV
Boundary range: 0.917 to 0.979
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data / Bkg
BDT score

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

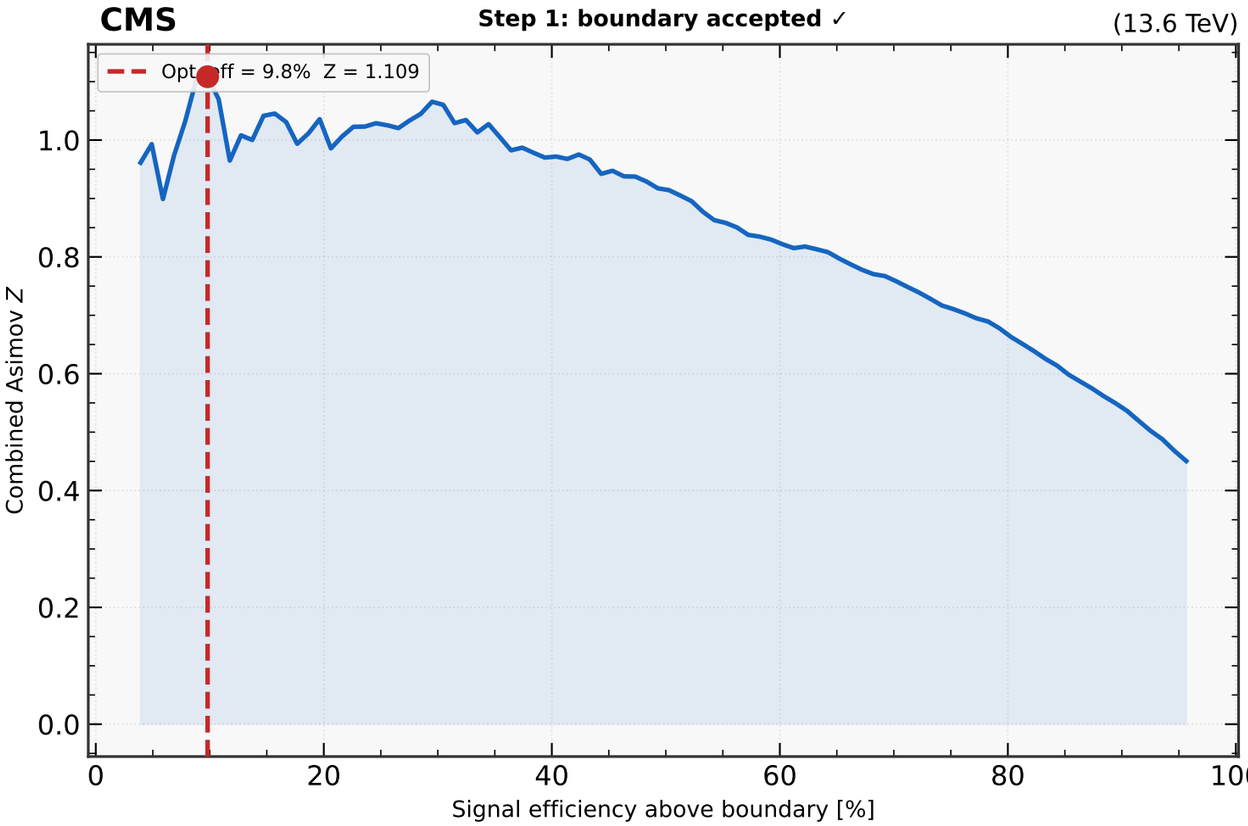
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.000, 0.025)	0.095	2525.40 ± 25.03	2319	0.00
[0.025, 0.050)	0.202	2299.53 ± 25.82	2326	0.01
[0.050, 0.075)	0.228	1660.35 ± 22.03	1683	0.01
[0.075, 0.100)	0.252	1274.05 ± 19.31	1355	0.02
[0.100, 0.125)	0.253	1002.27 ± 17.17	1039	0.03
[0.125, 0.150)	0.256	865.71 ± 16.12	891	0.03
[0.150, 0.175)	0.260	702.67 ± 14.32	728	0.04
[0.175, 0.200)	0.280	628.67 ± 13.98	611	0.04
[0.200, 0.225)	0.278	549.92 ± 12.78	571	0.05
[0.225, 0.250)	0.285	499.02 ± 12.57	490	0.06
[0.250, 0.275)	0.284	441.04 ± 11.54	437	0.06
[0.275, 0.300)	0.295	404.04 ± 11.20	420	0.07
[0.300, 0.325)	0.310	364.62 ± 10.64	356	0.08
[0.325, 0.350)	0.314	331.64 ± 10.10	347	0.09
[0.350, 0.375)	0.323	322.80 ± 9.87	303	0.10
[0.375, 0.400)	0.342	296.51 ± 9.47	325	0.12
[0.400, 0.425)	0.350	292.41 ± 9.76	266	0.12
[0.425, 0.450)	0.375	274.79 ± 9.27	285	0.14
[0.450, 0.475)	0.386	259.91 ± 8.93	263	0.15
[0.475, 0.500)	0.399	246.23 ± 8.90	243	0.16

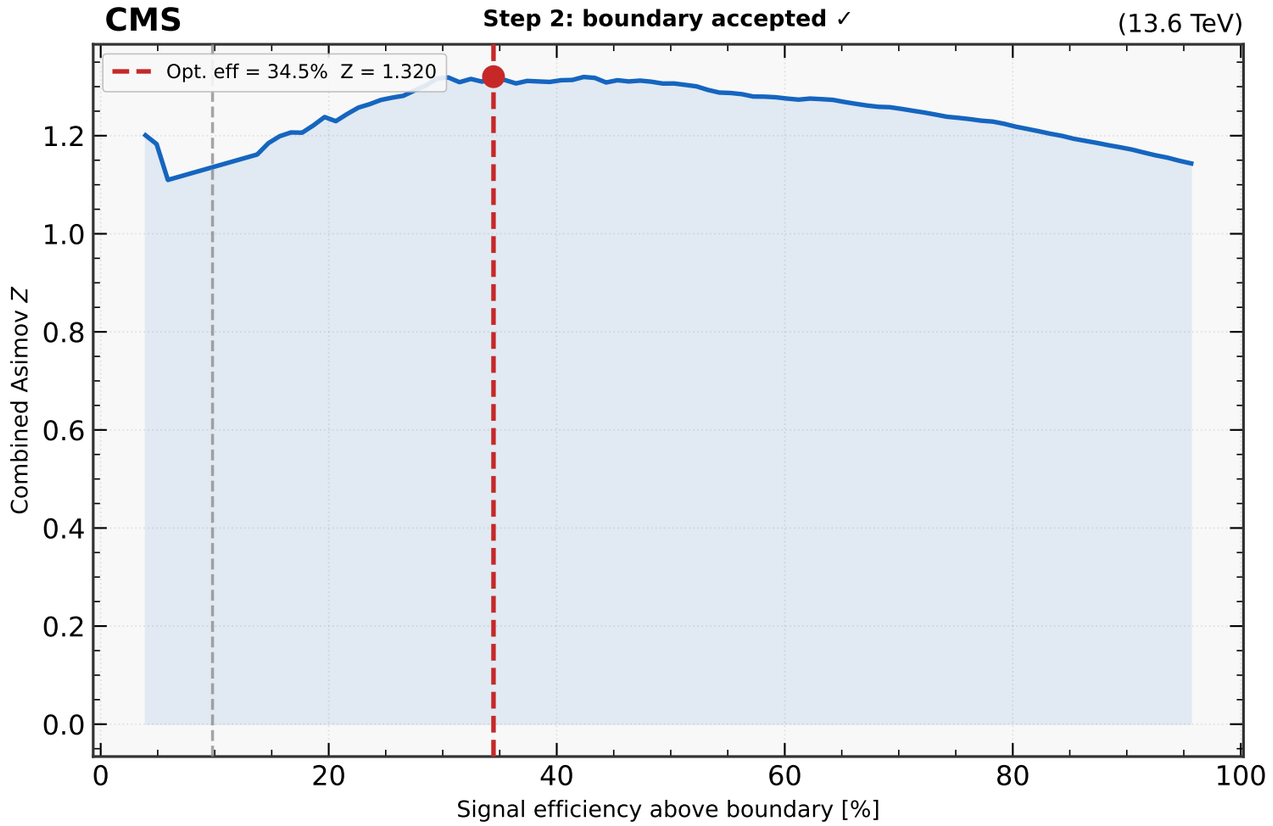
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

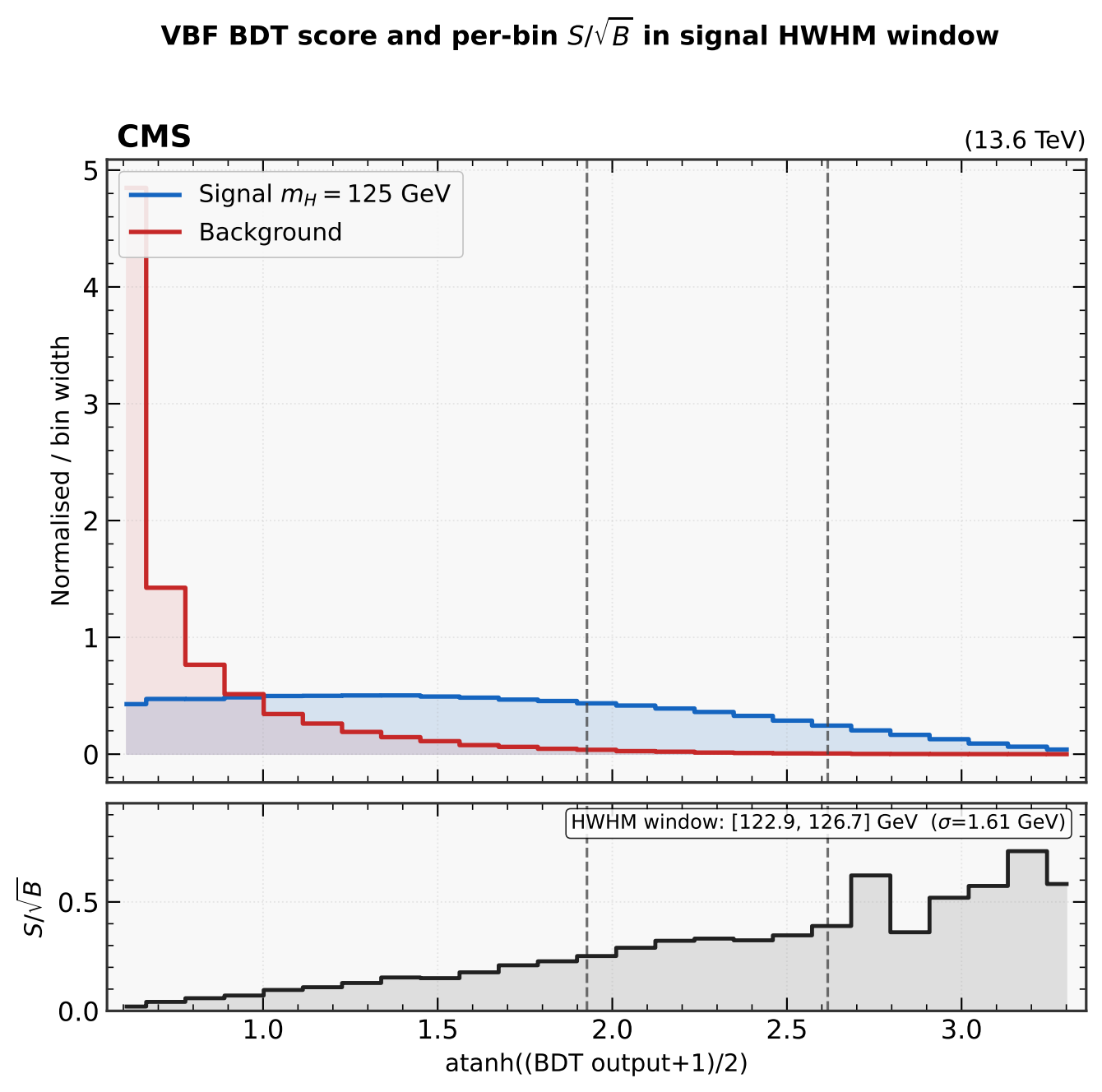
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.500, 0.525)	0.425	223.80 ± 8.32	218	0.19
[0.525, 0.550)	0.447	205.21 ± 7.92	207	0.22
[0.550, 0.575)	0.476	205.44 ± 7.52	207	0.23
[0.575, 0.600)	0.499	195.47 ± 7.77	182	0.25
[0.600, 0.625)	0.535	191.93 ± 7.59	181	0.28
[0.625, 0.650)	0.566	188.90 ± 8.18	191	0.30
[0.650, 0.675)	0.597	171.08 ± 6.85	179	0.35
[0.675, 0.700)	0.634	169.92 ± 7.06	172	0.37
[0.700, 0.725)	0.703	164.09 ± 7.22	151	0.43
[0.725, 0.750)	0.759	150.12 ± 6.67	154	0.50
[0.750, 0.775)	0.832	154.50 ± 6.71	163	0.54
[0.775, 0.800)	0.928	138.05 ± 6.22	154	0.67
[0.800, 0.825)	1.025	143.21 ± 6.60	156	0.71
[0.825, 0.850)	1.158	123.68 ± 6.05	139	0.93
[0.850, 0.875)	1.337	119.72 ± 5.65	139	1.10
[0.875, 0.900)	1.580	114.16 ± 5.62	110	1.37
[0.900, 0.925)	1.935	103.49 ± 5.27	118	1.84
[0.925, 0.950)	2.516	98.53 ± 5.34	118	2.49
[0.950, 0.975)	3.523	74.17 ± 4.30	99	4.53
[0.975, 1.000)	3.566	28.09 ± 2.22	40	11.27

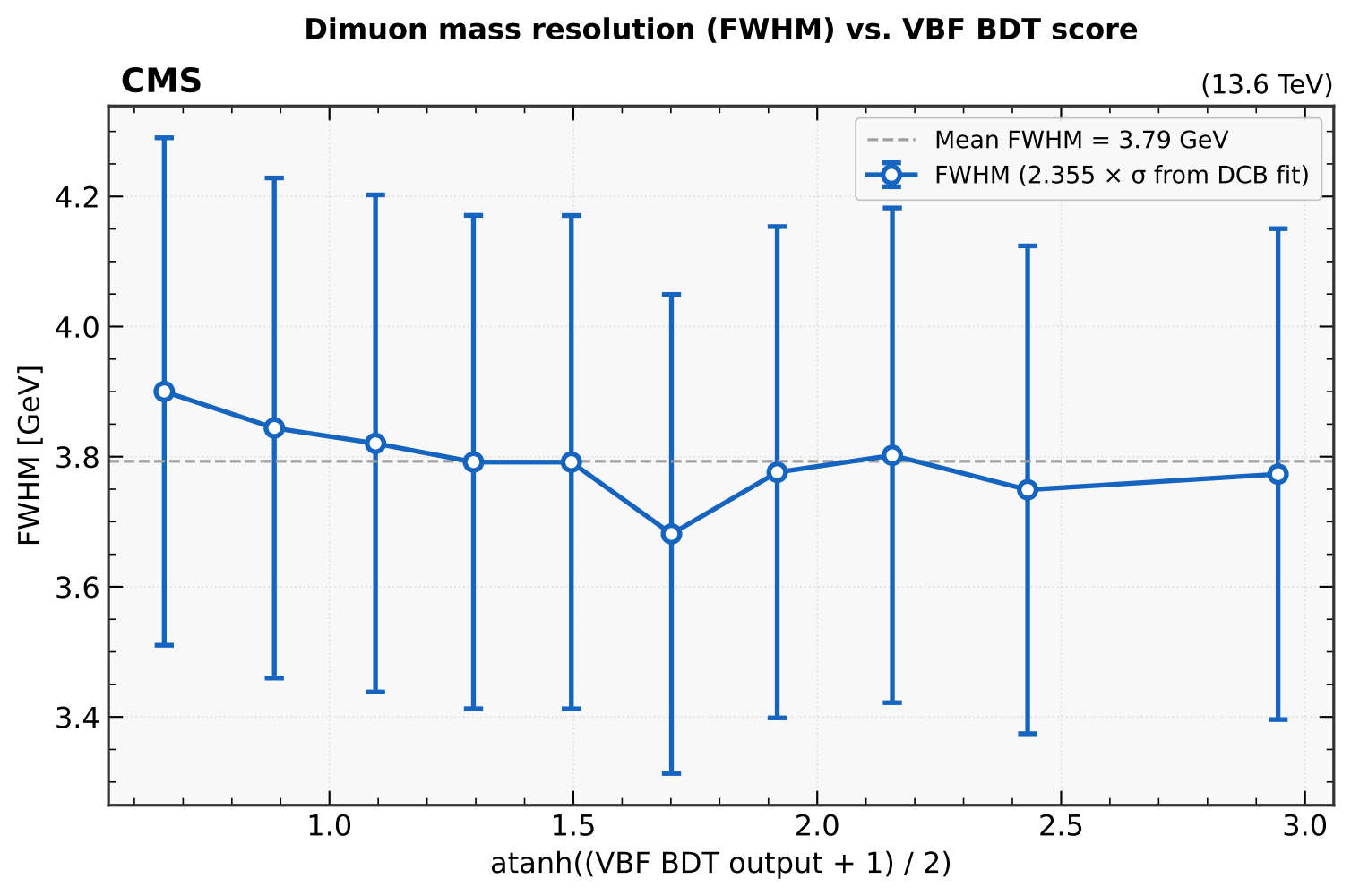

Combined significance vs. number of VBF BDT categories
CMS
(13.6 TeV)
Optimal: n=4, Z=1.3987
0.323
1.109
1.320
1.399
Combined Asimov Z [HWHM window]
Number of BDT categories
1
2
3
4
0.4
0.6
0.8
1.0
1.2
1.4

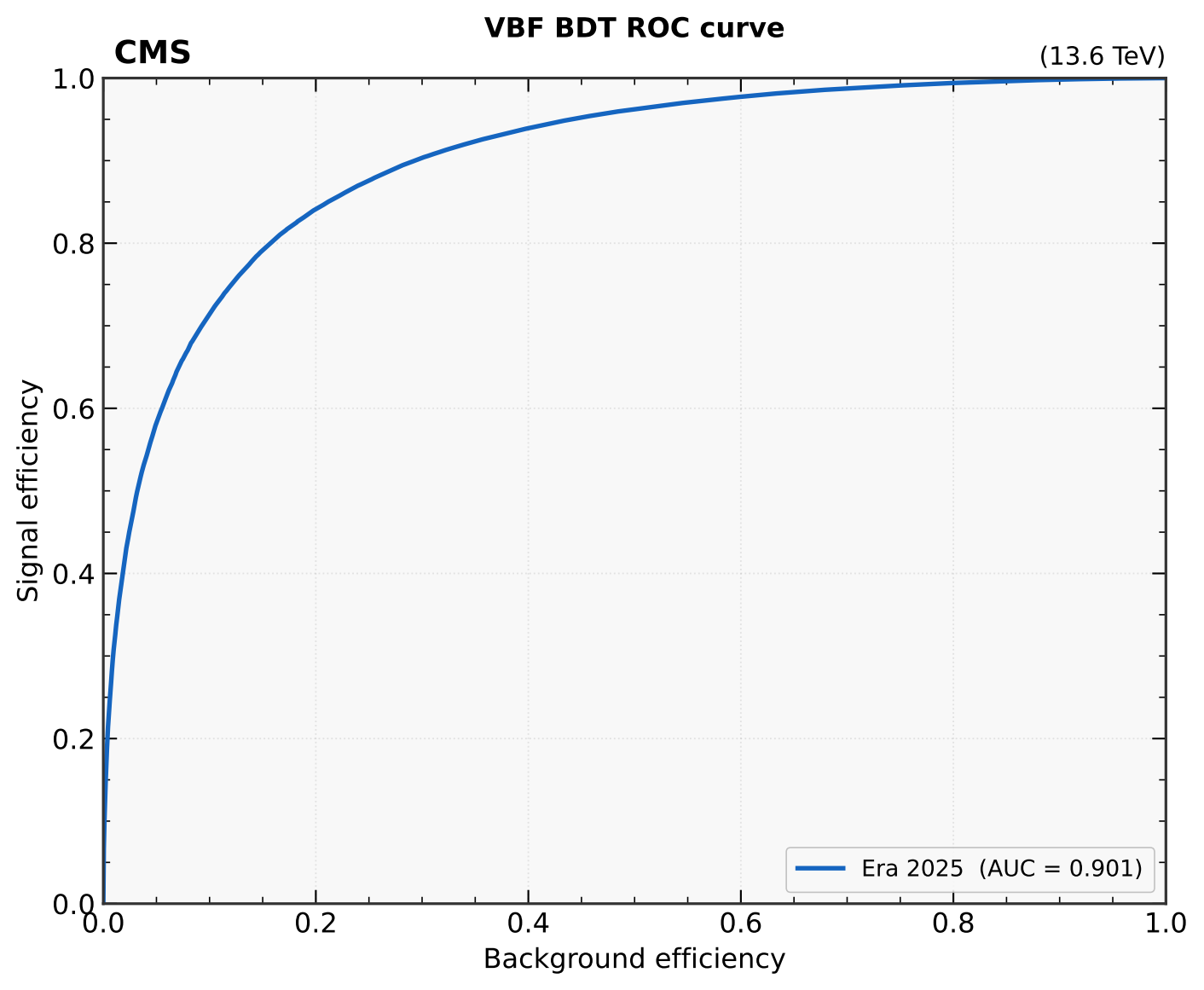
Iterative category boundary optimisation (AN Fig. 32 analog)

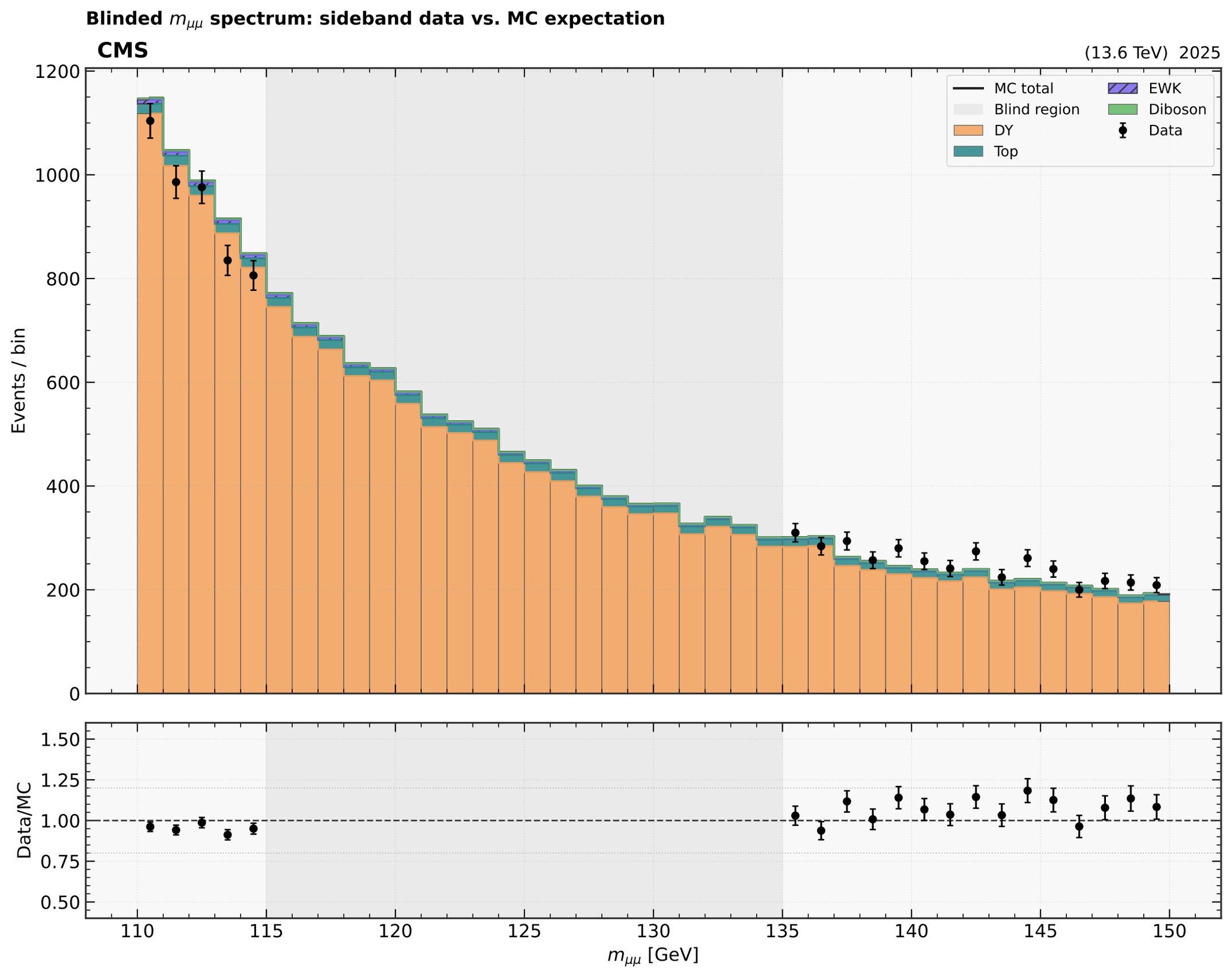


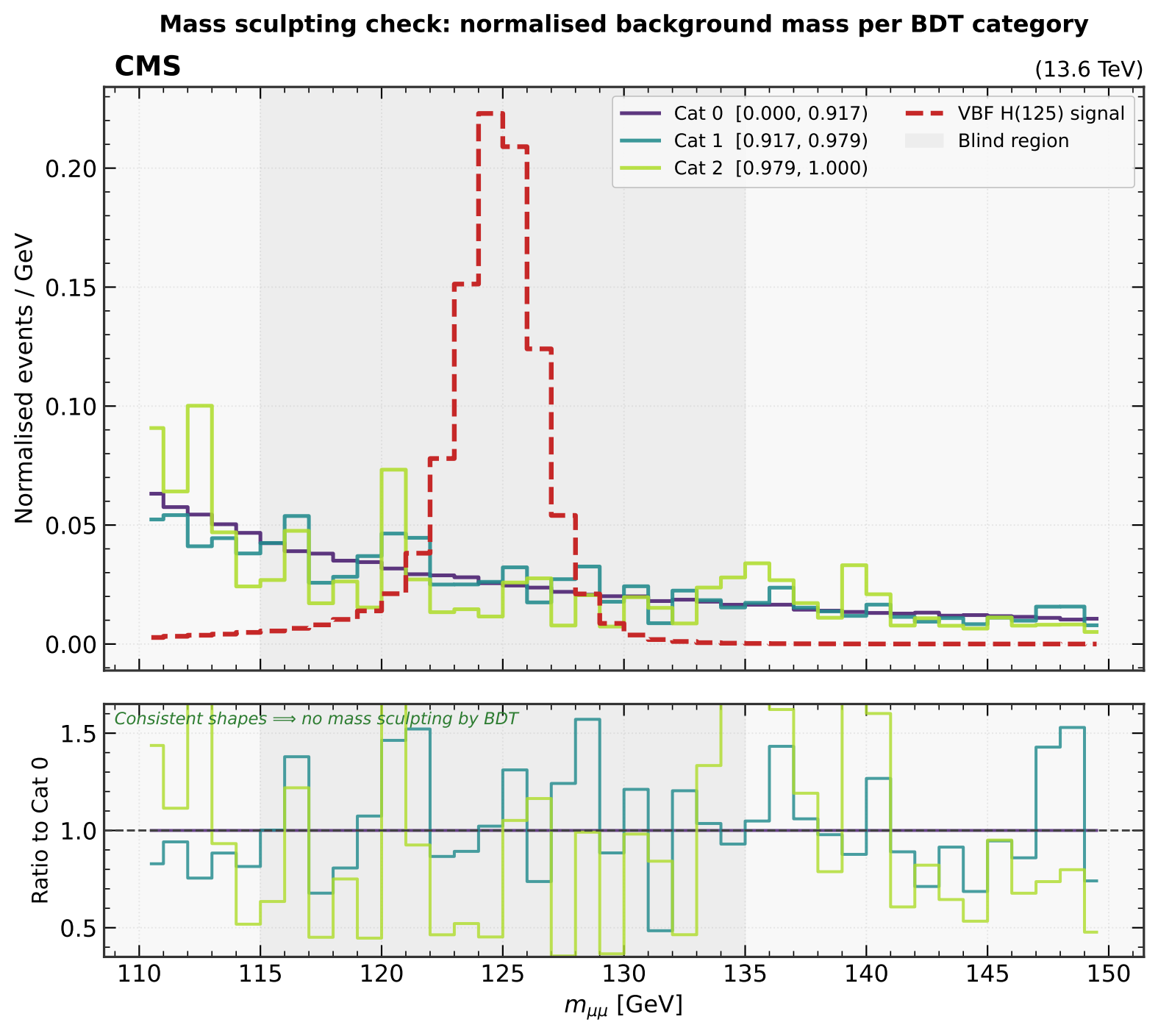



VBF BDT score and per-bin $S/\sqrt{B}$ in signal HWHM window
CMS
(13.6 TeV)
Signal $m_H = 125$ GeV
Background
Normalised / bin width
HWHM window: [122.9, 126.7] GeV (σ=1.61 GeV)
$S/\sqrt{B}$
atanh((BDT output+1)/2)

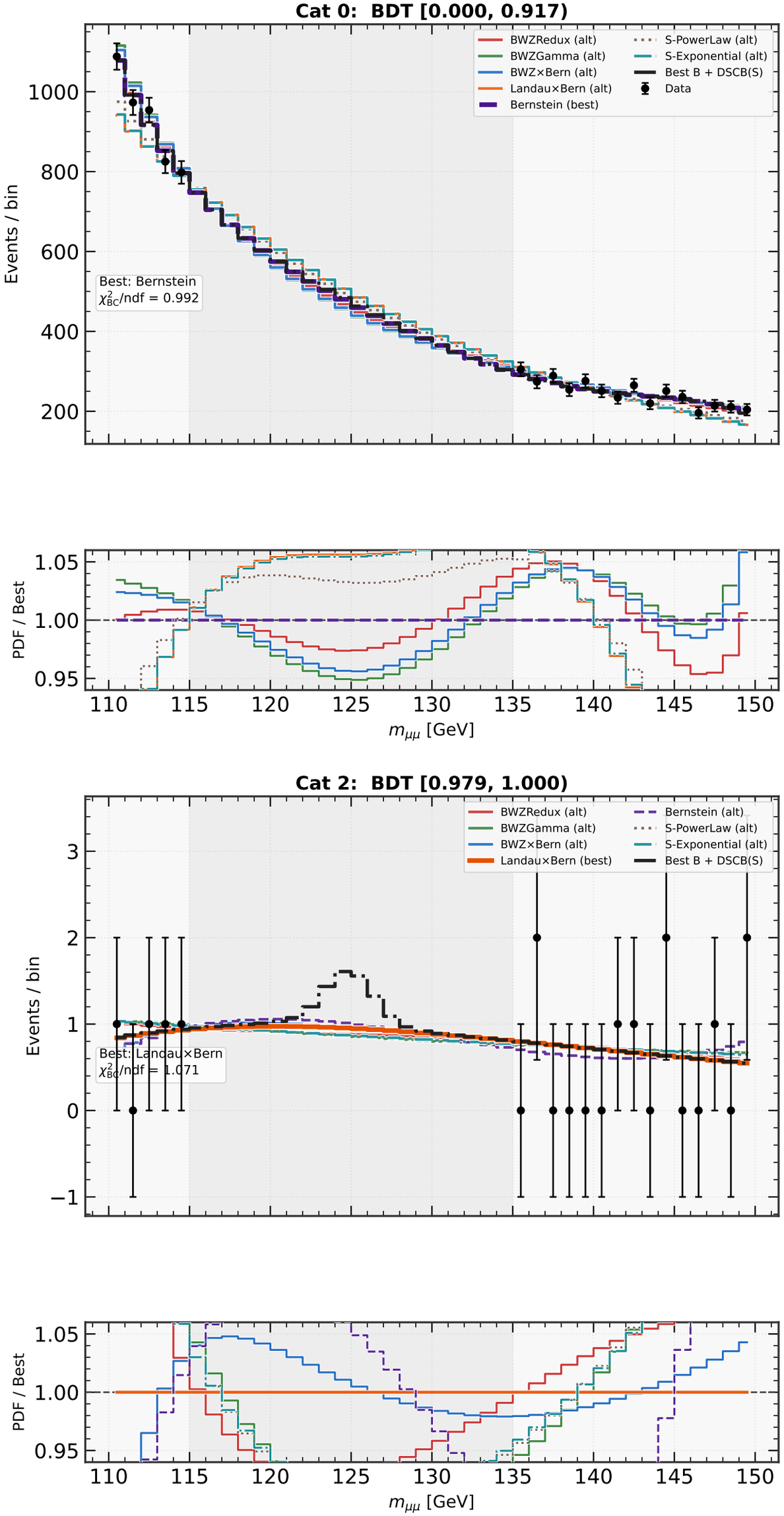

Dimuon mass resolution (FWHM) vs. VBF BDT score
CMS
(13.6 TeV)
Mean FWHM = 3.79 GeV
FWHM (2.355 × σ from DCB fit)
FWHM [GeV]
4.2
4.0
3.8
3.6
3.4
1.0
1.5
2.0
2.5
3.0
atanh((VBF BDT output + 1) / 2)

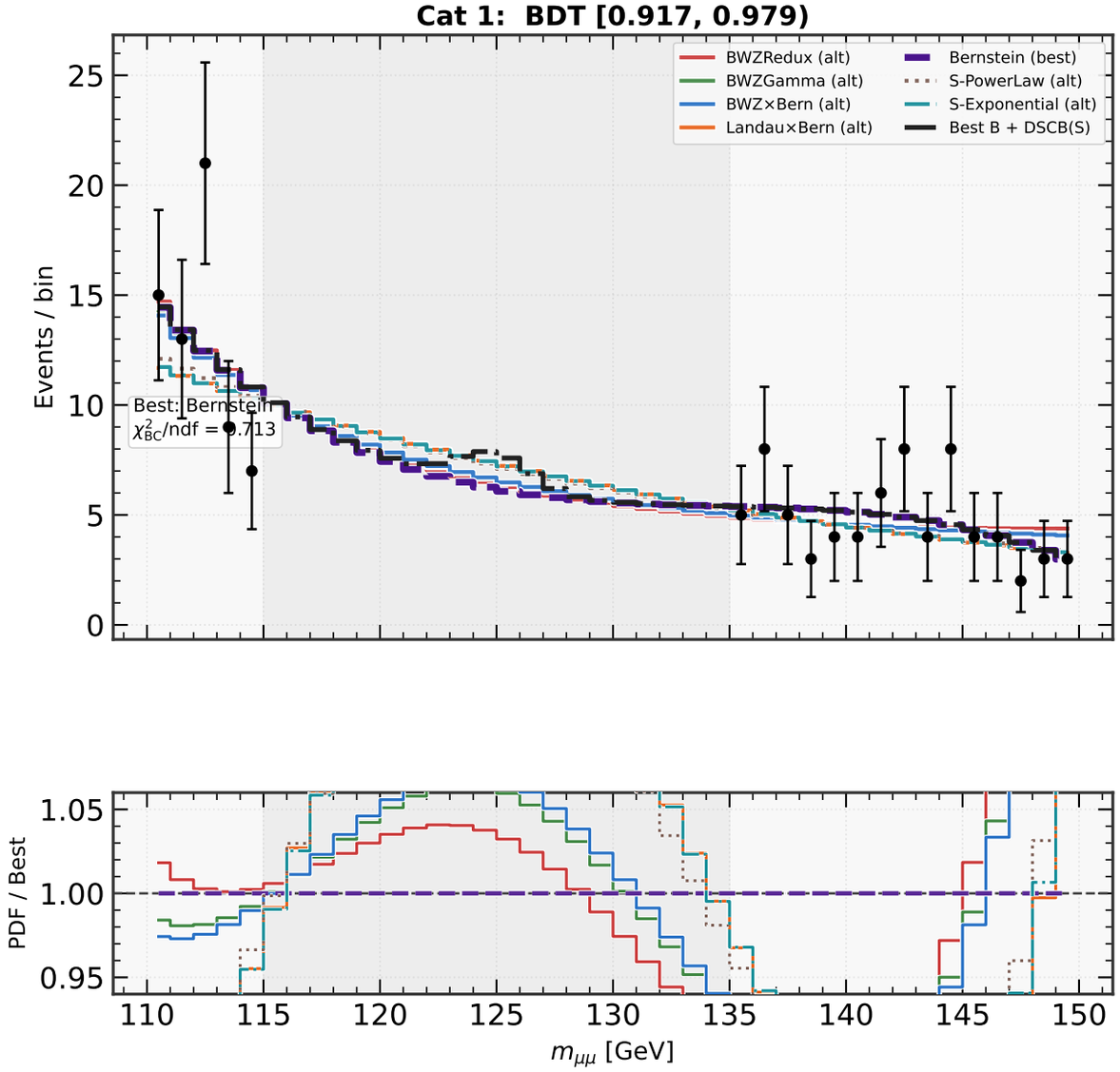

VBF BDT ROC curve
CMS
(13.6 TeV)
Signal efficiency
Background efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Era 2025 (AUC = 0.901)


Blinded $m_{\mu\mu}$ spectrum: sideband data vs. MC expectation
CMS
(13.6 TeV) 2025
MC total
Blind region
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
$m_{\mu\mu}$ [GeV]


Mass sculpting check: normalised background mass per BDT category
CMS
(13.6 TeV)
Cat 0 [0.000, 0.917)
Cat 1 [0.917, 0.979)
Cat 2 [0.979, 1.000)
VBF H(125) signal
Blind region
Normalised events / GeV
0.00
0.05
0.10
0.15
0.20
Consistent shapes ⟹ no mass sculpting by BDT
Ratio to Cat 0
0.5
1.0
1.5
110
115
120
125
130
135
140
145
150
$m_{\mu\mu}$ [GeV]

Per-category dimuon mass: background model fits

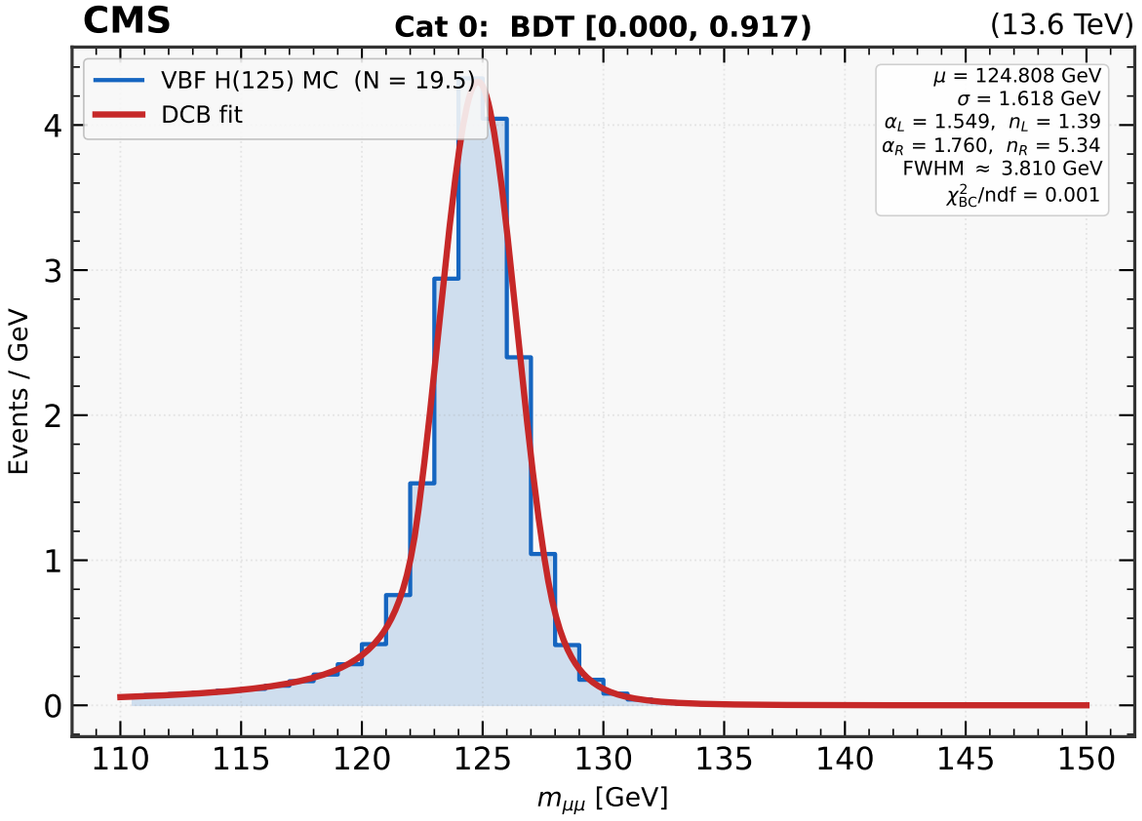


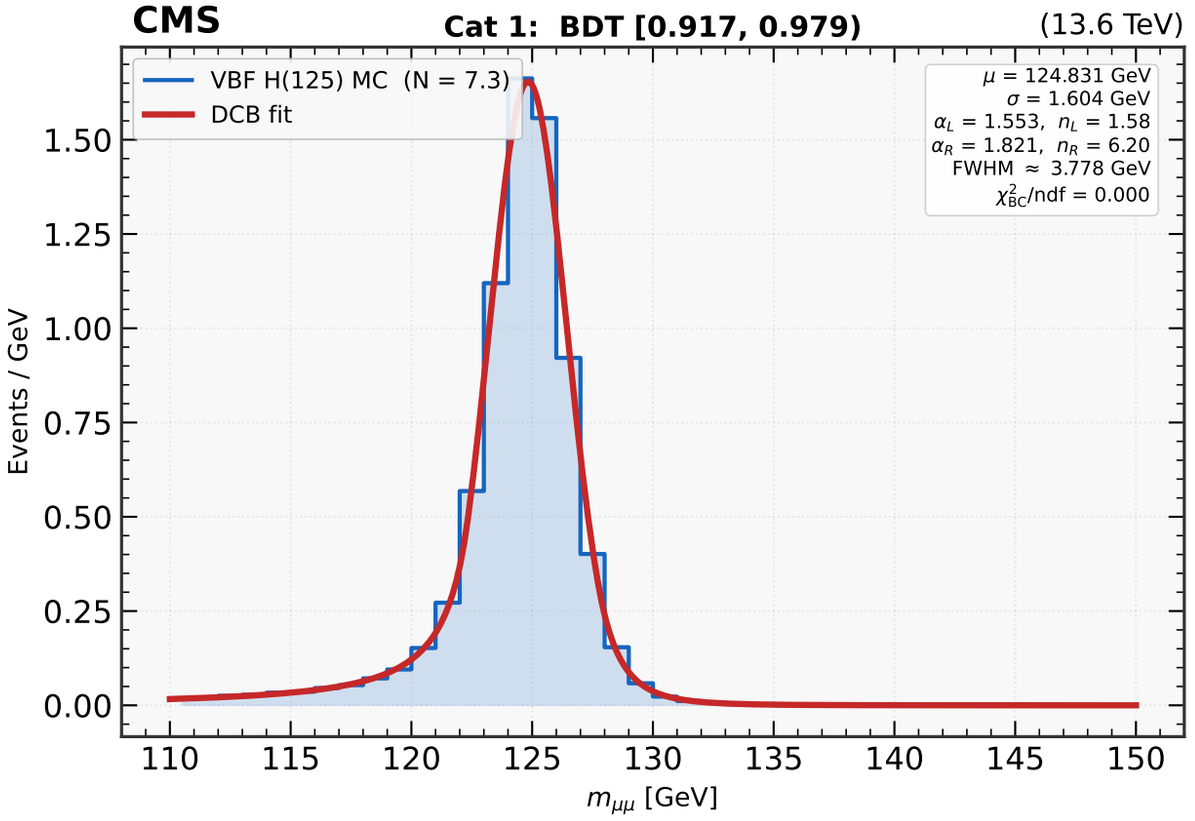


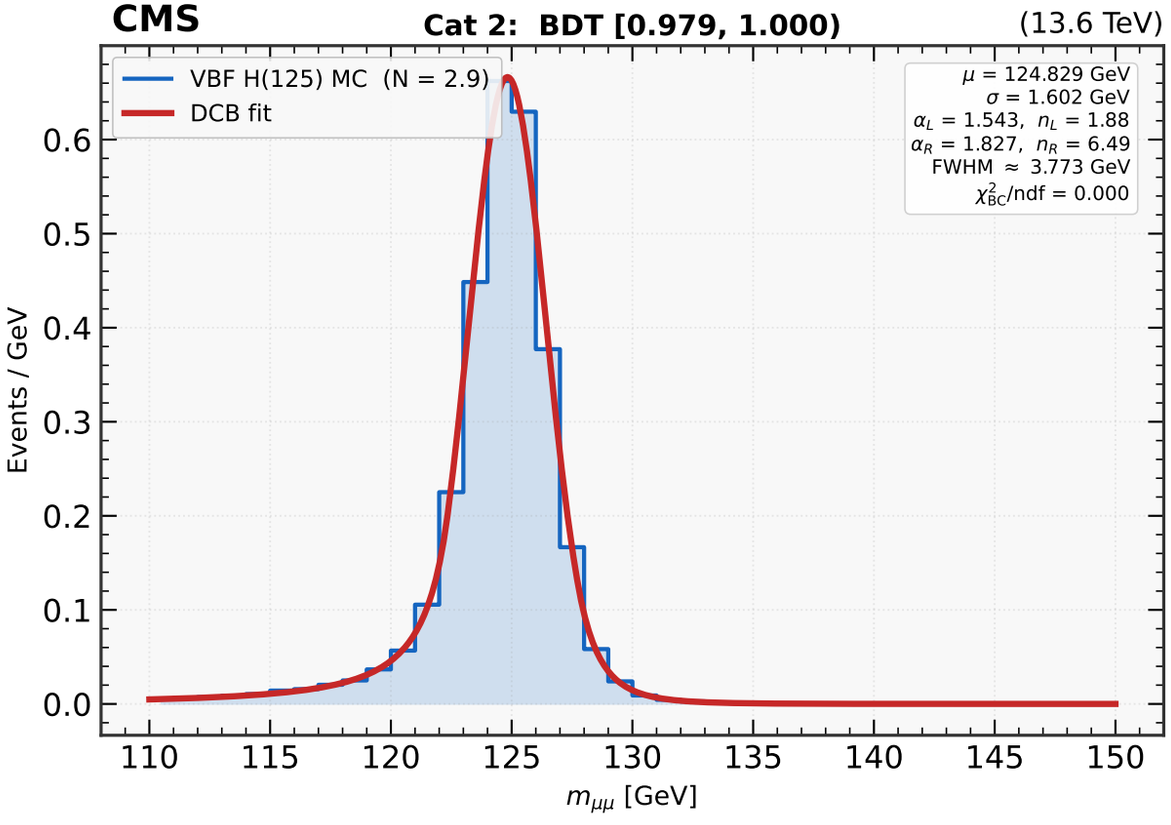
Per-category background model fit comparison (χ²_BC/ndf, ndf)

Category	BDT range	Best model	Best χ²_BC/ndf	BWZRedux	BWZGamma	BWZ×Bern	Landau×Bern	Bernstein	S-PowerLaw	S-Exponential
0	[0.000, 0.917)	Bernstein	0.992	1.119 (37)	1.383 (37)	1.284 (36)	3.543 (37)	0.992 (33)	2.304 (37)	3.439 (37)
1	[0.917, 0.979)	Bernstein	0.713	0.748 (37)	0.752 (37)	0.776 (36)	0.891 (37)	0.713 (35)	0.858 (37)	0.884 (37)
2	[0.979, 1.000)	Landau×Bern	1.071	1.081 (37)	1.075 (37)	1.094 (36)	1.071 (37)	1.101 (35)	1.076 (37)	1.076 (37)

VBF H(125) signal DCB extraction per BDT category

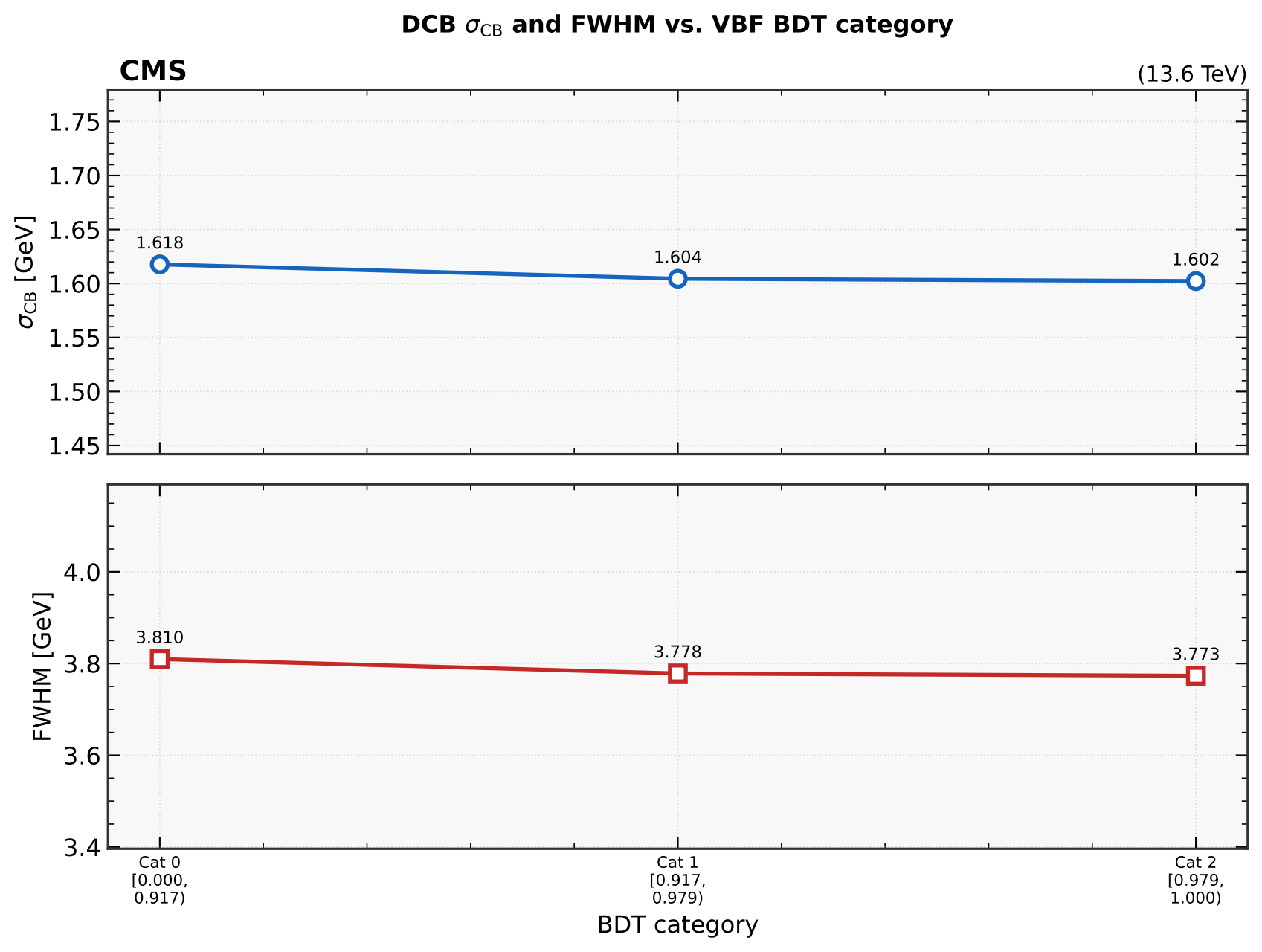


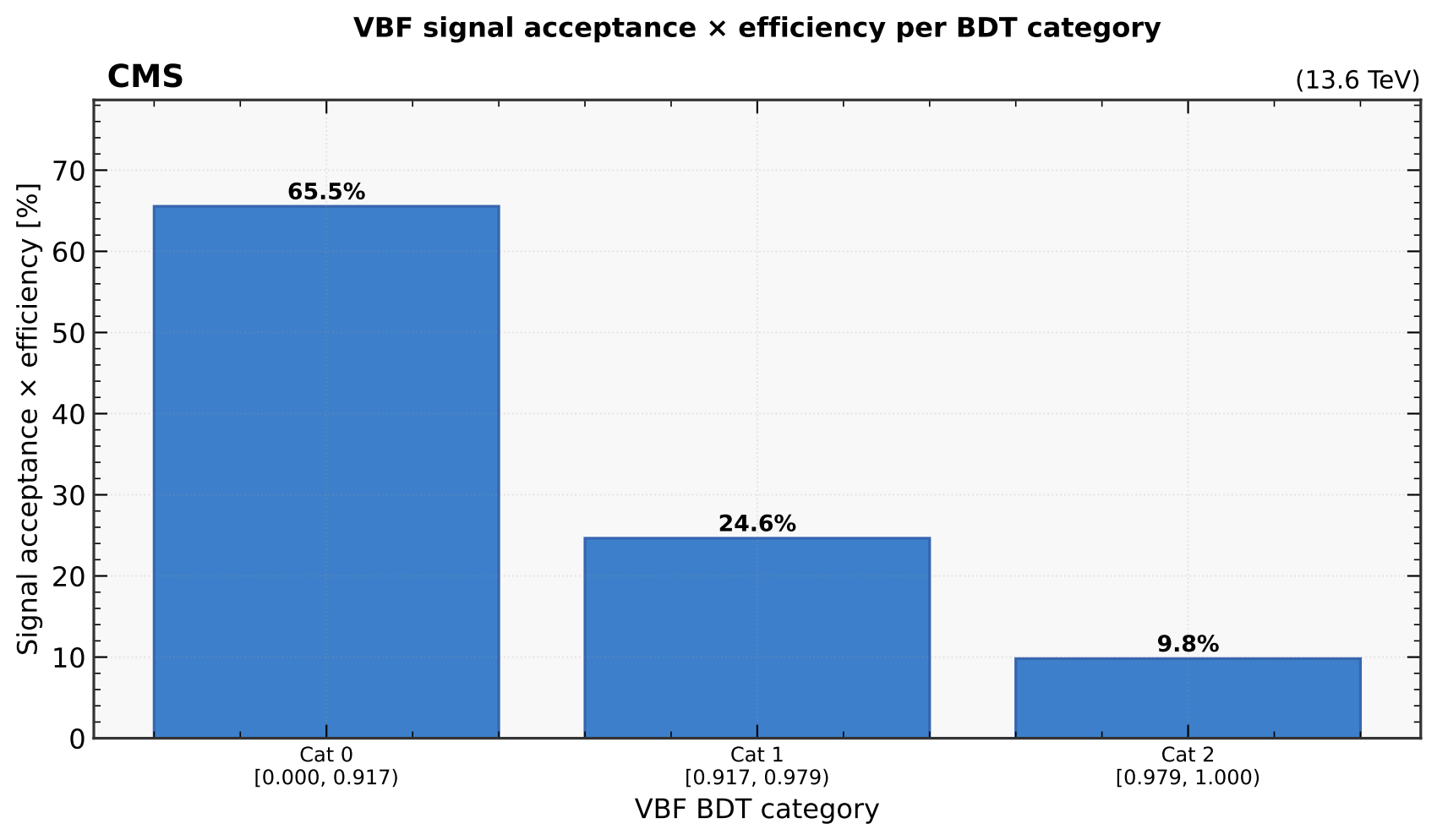




VBF signal DCB fit parameters per category

Category	BDT range	N_sig	Mean [GeV]	Sigma [GeV]	alpha_L	n_L	alpha_R	n_R	FWHM [GeV]	Asimov Z	χ^2_BC/ndf
0	[0.000, 0.917)	19.54	124.8081	1.6177	1.5486	1.394	1.7603	5.344	3.810	0.211	0.001
1	[0.917, 0.979)	7.35	124.8306	1.6044	1.5529	1.583	1.8210	6.204	3.778	0.744	0.000
2	[0.979, 1.000)	2.92	124.8288	1.6022	1.5426	1.879	1.8266	6.486	3.773	1.075	0.000


DCB σ_{CB} and FWHM vs. VBF BDT category
CMS
(13.6 TeV)
σ_{CB} [GeV]
1.75
1.70
1.65
1.60
1.55
1.50
1.45
1.618
1.604
1.602
FWHM [GeV]
4.0
3.8
3.6
3.4
3.810
3.778
3.773
Cat 0 [0.000, 0.917)
Cat 1 [0.917, 0.979)
Cat 2 [0.979, 1.000)
BDT category


VBF signal acceptance × efficiency per BDT category
CMS
(13.6 TeV)
Signal acceptance × efficiency [%]
0
10
20
30
40
50
60
70
65.5%
24.6%
9.8%
Cat 0
[0.000, 0.917)
Cat 1
[0.917, 0.979)
Cat 2
[0.979, 1.000)
VBF BDT category

Per-category yield and sensitivity summary

Category	BDT range	S_full	B_full	S_HWHM	B_HWHM	HWHM window [GeV]	S/√B	Asimov Z
0	[0.000, 0.917)	19.54	40802.08	13.393	4014.275	[122.9, 126.7]	0.211	0.211
1	[0.917, 0.979)	7.35	480.86	5.111	45.537	[122.9, 126.7]	0.757	0.744
2	[0.979, 1.000)	2.92	43.50	2.058	3.043	[122.9, 126.7]	1.180	1.075

Category summary: HWHM-windowed signal and background yields

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	S/√B [HWHM]	Asimov Z (cat) [HWHM]	B_HWHM
0	[0.000, 0.917)	8318	18990.63 ± 111.98	0.438	40802.08 ± 164.37	0.2114	0.2113	4014.275
1	[0.917, 0.979)	136	206.23 ± 10.60	0.659	480.86 ± 17.14	0.7574	0.7439	45.537
2	[0.979, 1.000)	13	23.59 ± 3.26	0.551	43.50 ± 4.07	1.1799	1.0745	3.043
Combined	all						1.3239	

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total	total_err2
0	[0.000, 0.917)	39028.96	243.31	1377.98	151.84	40802.08	27016.80
1	[0.917, 0.979)	398.30	68.65	11.96	1.95	480.86	293.92
2	[0.979, 1.000)	23.08	19.46	0.85	0.10	43.50	16.58